Shuxrat Mahmadaliev



Backend Developer · **4.7 years of experience** · Node.js / TypeScript · Python · Rust

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github.com/Shuxrat-dat | Portfolio: https://portfolio-gules-pi-0zfovrcvtz.vercel.app/

ABOUT

Backend developer with 4.7 years of experience in web development and project work; primary stack — Node.js/TypeScript, PostgreSQL, and React/Next.js. I design REST APIs, data models, and server-side business logic, and build payment services, multi-tenant platforms, and AI tools. In PayForge I implemented payment state machines, integer money accounting, a ledger, idempotency, queues, and signed webhooks. I work with PostgreSQL/Prisma, Redis, RabbitMQ/BullMQ, Docker, GitHub Actions, and automated tests. I can own a backend feature from the data schema and API contract through background processing, tests, and a reproducible environment; when needed I also deliver the UI in React/Next.js. Looking for a Backend Developer role on a team that values API reliability, security, and maintainability.

SKILLS

Languages	TypeScript, JavaScript, Python, Rust, SQL, HTML, CSS, MQL5 (integration)
Backend & API	Node.js, Express, NestJS, FastAPI, Axum, REST API, OpenAPI/Swagger, Socket.io
Databases	PostgreSQL, Prisma ORM/Migrate, SQLx, Redis, LanceDB, IndexedDB
Frontend	React, Next.js, Vite, Tailwind CSS, TanStack Query, Zustand, Framer Motion
Bots & AI	Telegram bots, aiogram, RAG, Ollama, OpenAI API, bge-m3, LanceDB, faster-whisper
Testing & quality	Jest, Supertest, pytest, Vitest, Playwright; unit, integration, and E2E tests
DevOps & tools	Linux, Docker, Docker Compose, Nginx, GitHub Actions, Turborepo, pnpm, npm, uv, Git
Architecture	REST API, modular architecture, multitenancy, RBAC/ACL, JWT, API keys, idempotency, queues, webhooks, audit
Additional	MySQL, PHP, Java, Lua; web programming, database design, internet project management

KEY PROJECTS · PERSONAL ENGINEERING PRACTICE

PayForge

github.com/Shuxrat-dat/PayForge

2026 · Backend payment-gateway sandbox · contribution: architecture, API, tests, infrastructure

Payment gateway sandbox with no real money movement: 21 feature modules and 32 Prisma models for payment intents, wallets/ledger, subscriptions, invoices, refunds, disputes, hosted checkout, and webhooks.

- Implemented BigInt money, guarded state transitions, a ledger under `SELECT FOR UPDATE`, Idempotency-Key, and Redis-based fraud scoring.
- Built a dedicated worker: RabbitMQ retry/DLQ, clearing, dunning, webhook retries; JWT refresh rotation with reuse detection, API keys, and HMAC-signed webhooks.
- Covered the project with 65 Jest/Supertest tests; set up Docker Compose, GitHub Actions, and OpenAPI 3 for about 74 API paths.

Stack: TypeScript, Node.js, Express, PostgreSQL, Prisma, Redis, RabbitMQ, Jest, Docker, GitHub Actions.

Value: demonstrates work with a money domain, concurrent updates, asynchronous failure handling, and verifiable API contracts.

AI Trading Agent

github.com/Shuxrat-dat/Ai-Agent

2026 · Local AI/automation MVP · contribution: services, integrations, safety, tests

Python monorepo of 11 services (~6K LOC): Obsidian RAG, market analysis, Telegram, voice operator, PyQt6 HUD, and MetaTrader 5 control via an MQL5 JSON bridge.

- Unified interfaces through AgentBrain; implemented bge-m3 + LanceDB hybrid search and an analysis engine where code computes numeric facts and the LLM only interprets them.
- Added a numeric validator and deterministic fallback, level monitor, discipline journal, healthcheck, and demo-only trade execution with four safety gates.
- Created 133 pytest tests and captured key decisions in 4 ADRs; supported local Ollama and optional OpenAI.

Stack: Python 3.12, FastAPI, Ollama/OpenAI, LanceDB, PyArrow, aiogram, MetaTrader5, MQL5, faster-whisper, PyQt6, pytest.

Value: shows multi-service architecture, AI integration with real tools, and safety-by-design rather than a thin LLM wrapper.

KEY PROJECTS · CONTINUED

CompanyOS

github.com/Shuxrat-dat/companyos

2026 · Early multi-tenant SaaS platform MVP · contribution: full-stack architecture

Corporate platform monorepo: NestJS API, Next.js web, BullMQ worker, and cron scheduler; 26 API modules, 96 Prisma models, and about 31 UI pages.

- Implemented JWT access/refresh rotation, OAuth strategies, API keys, tenant guard, RBAC + resource ACL, audit logging, and rate limiting.
- Built projects/tasks, documents, calendar, Socket.io chat, MinIO/S3 files, and background queues for email, AI, analytics, and webhooks.
- Assembled pnpm/Turborepo, Docker Compose, Nginx, and GitHub Actions for CI and container builds; external billing/AI integrations work when env is configured.

Stack: NestJS, Next.js 15, TypeScript, PostgreSQL, Prisma, Redis, BullMQ, Socket.io, MinIO, Docker, Turborepo.

Value: demonstrates multi-tenant SaaS design from the data schema and security guards through the web client and background processing; not positioned as a mature production product.

SwitchLab / Switchyard

github.com/Shuxrat-dat/SwitchLab

2026 · Full-stack e-commerce demo · contribution: Rust API, React UI, testing, CI

Mechanical keyboard store with catalog, cart, checkout, user account, and admin panel; payment is intentionally simulated.

- Built a Rust/Axum API with PostgreSQL/SQLx, Argon2 sessions, CSRF/Origin protection, roles, and OpenAPI/Swagger.
- Secured checkout with a transaction and stock locking via `FOR UPDATE`; implemented React/Vite storefront and admin flows.
- Set up Docker Compose and GitHub Actions: rustfmt, clippy, backend integration, Vitest, production build, and Playwright E2E.

Stack: Rust, Axum, Tokio, SQLx, PostgreSQL, React 19, TypeScript, Vite, TanStack Query, Playwright, Docker.

Value: confirms ability to work beyond the primary Node.js stack, design a transactional backend, and take a flow through to CI/E2E.

ADDITIONAL PROJECTS

TasteBase · GitHub

Full-stack recipe platform: Next.js 15, Auth.js, Prisma/PostgreSQL, CRUD, search, filters, pagination, favorites, profile, and ownership checks. Demonstrates feature-based structure, Zod, and TanStack Query practice.

Trading Journal · GitHub

Next.js 14 MVP for trade logging: CRUD, psychology and risk fields, NextAuth/Prisma, win rate / profit factor / drawdown stats, and an equity curve on Recharts.

MassUp (Massa-Nabor) · GitHub

Offline-first fitness PWA on Next.js 15: 12-week program, 42 exercises, trackers, nutrition, Service Worker, Zustand + IndexedDB, and JSON backup. Fully client-side.

Portfolio · https://portfolio-gules-pi-0zfovrcvtz.vercel.app/

Personal site on Next.js 15 / React 19 / Tailwind CSS 4: data-driven content, responsive UI, Framer Motion, Lenis, image optimization, and reduced-motion support.

VERIFIED ENGINEERING OUTCOMES

- 8 public repositories: backend, full-stack, AI tooling, and PWA work in TypeScript, Python, and Rust.
- 198 automated tests across PayForge and AI Trading Agent; unit/integration/E2E pipeline in SwitchLab.
- REST APIs with OpenAPI, idempotency, row locking, HMAC webhooks, RBAC/ACL, tenant isolation, and audit trail.
- Asynchronous workflows on RabbitMQ and BullMQ with retry/DLQ, schedulers, and Redis locks.
- Reproducible environments via Docker Compose, Nginx, and GitHub Actions CI.
- Architecture docs and ADRs for AI/RAG, analysis engine, and platform-control decisions.

EDUCATION

Self-directed engineering study and project practice: backend, databases, testing, distributed workflows, and AI integrations.

LANGUAGES

Russian
English

ADDITIONAL

Public GitHub, Git, architectural ADRs, local AI models. Open to remote work.